PROPORTIONS TOOLBOX

new 3.2–3.13 • old 6.1–6.11

ZONE A · ONE SAMPLE / TWO INDEPENDENT SAMPLES

ONE PROPORTION · CONFIDENCE INTERVAL

$$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

ONE PROPORTION · TEST

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

$H_0 : p = p_0$. Use the null value p_0 in the denominator.

TRUE SD · WHEN POPULATION PROPORTIONS ARE KNOWN

$$\sigma_{\hat{p}_1-\hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Standard error ($s_{\text{statistic}}$, or SE) estimates the true SD of a sampling distribution using sample data.

TWO PROPORTIONS · CONFIDENCE INTERVAL

$$s_{\hat{p}_1-\hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

TWO PROPORTIONS · TEST OF $H_0 : p_1 = p_2$

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

$$\text{SE}_{\text{pooled}} = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1-\hat{p}_c)(1/n_1 + 1/n_2)}}$$

x_i: successes; n_i: sample size; $\hat{p}_i = x_i/n_i$.

ZONE B · CONDITIONS + WHICH PROPORTION TO USE

1 · **Random:** random sample or randomized experiment; separate groups independent.
2 · **10%:** when sampling without replacement, $n < 0.10N$ in each population.
3 · **Large Counts:** at least 10 expected successes and 10 expected failures.

CI: use the sample proportion
$n\hat{p} \geq 10, \quad n(1-\hat{p}) \geq 10$
Two groups: check each $\hat{p}_i$ with its own n_i.

Test: use the null model
$np_0 \geq 10, \quad n(1-p_0) \geq 10$
Two groups: use $\hat{p}_c$; check both $n_i\hat{p}_c$ and $n_i(1-\hat{p}_c)$.

ZONE C · SAY IT IN WORDS

"We are ____% confident the true proportion of ____________________ is between ______ and ______."

For two groups, name the difference $p_1 - p_2$ in context and keep the subtraction order.

ZONE D · WATCH OUT

CI: estimate each unknown proportion with its own $\hat{p}$.
One-sample test: use p_0. **Two-sample test:** pool under $p_1 = p_2$.
Keep the CI unpooled. Its groups may have different true proportions.

see WS old u6_l1–11